

Munkres 9

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1 Introduction

Let $p : E \rightarrow B$ be a covering map. Then p is an open map.

Let $U(x) \subset E$ be open. Let $y = p(x)$ and pick $V(y)$ such that V is evenly covered by p . Since $x \in p^{-1}(V)$, pick $x \in V_\alpha$. Then $(p|_{V_\alpha})(V_\alpha \cap U) \subset B$ is open (since U, V are open, $p|_{V_\alpha}$ is a homeomorphism). Also, $y \in (p|_{V_\alpha})(V_\alpha \cap U) \subset p(V_\alpha \cap U) \subset p(U)$, so $p(U)$ is open.

Let X have the discrete topology. Then $\pi_1 : Y \times X \rightarrow Y$ is a covering map. $\pi_1^{-1}(Y) = \bigcup \{Y \times \{x\} : x \in X\}$, where each slice V_x is open and homeomorphic to Y under restriction of π_1 .

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be linear. Then f is a closed map. Let $f(x) = ax + b$. If $a = 0$, then f is closed. Else, $a \neq 0$ and f is a homeomorphism and in particular closed.

If $p : E \rightarrow B$ is a covering map and B is Hausdorff, so is E . Let $e_1 \neq e_2 \in E$. If $p(e_1) \neq p(e_2)$, we can pick $U(p(e_1)), V(p(e_2))$ open disjoint, for which $p^{-1}(U), p^{-1}(V)$ are disjoint open neighborhoods of e_1, e_2 , respectively.

Else, $p(e_1) = p(e_2) = b$. Let $U(b)$ be evenly covered by p and $\{V_\alpha\}$ be a partition of $p^{-1}(U)$ into slices. If $e_1 \in V_\alpha, e_2 \in V_\beta$ for $\alpha \neq \beta$, we're done. Else, $e_1, e_2 \in V_\alpha$. Since $V_\alpha \cong U$ and U is Hausdorff, we may pick open disjoint neighborhoods of e_1, e_2 in V_α , and therefore E .

Let $p : E \rightarrow B$ be a covering map, $f : I \rightarrow B$ a path starting at b_0 , and $e_0 \in E$ such that $p(e_0) = b_0$. Then f lifts uniquely to $\tilde{f} : I \rightarrow E$.

I first show existence. Let $\{U\}$ be an open cover of B with evenly covered sets by p . Since f is continuous, $\{f^{-1}(U)\}$ is an open cover of I . By Lebesgue number lemma, since I is compact metrizable, pick a subdivision $0 = s_1 < s_2 < s_3 < \dots < s_n = 1$ such that $[s_i, s_{i+1}] \subset f^{-1}(U)$ for some U , for all $i = 0, 1, 2, \dots, n-1$. Then $f([s_i, s_{i+1}]) \subset U$. Fix a U_i for each i .

I claim there exists a function $\tilde{f}_k : [0, s_k] \rightarrow E$ such that $\tilde{f}_k(0) = e_0$ and $p \circ \tilde{f}_k = f$ on $[0, s_k]$ for $k \in [n]$. I proceed by induction.

For $k = 1$, since $b_0 \in U_1$, $e_0 \in p^{-1}(\{b_0\}) \subset p^{-1}(U_1)$, so we may pick a slice V_1 of $p^{-1}(U_1)$ containing e_0 . Let $\tilde{f}_1 : [0, s_1] \rightarrow V_1$ be given by $(p|_{V_1})^{-1} \circ f$, well-defined continuous since $p|_{V_1}$ is a homeomorphism and $f([0, s_1]) \subset U_1$. We have $p \circ \tilde{f}_1 = f$ on $[0, s_1]$. Further, $(p|_{V_1})(\tilde{f}_1(0)) = f(0) = b_0 = (p|_{V_1})(e_0)$, then $\tilde{f}_1(0) = e_0$ since $p|_{V_1}$ is injective.

Assume the result for $k < n$. Define $\tilde{f}_{k+1}$ as follows. Since $p \circ \tilde{f}_k(s_k) = f(s_k) \in U_{k+1}$, we have $\tilde{f}_k(s_k) \in p^{-1}(f(s_k)) \subset p^{-1}(U_{k+1})$. Pick the slice V_{k+1} of $p^{-1}(U_{k+1})$ containing $\tilde{f}_k(s_k)$. Define $\tilde{f}_{k+1} : [0, s_{k+1}] \rightarrow \bigcup_{i \leq k+1} V_i$ by
$$\tilde{f}_{k+1}(s) = \begin{cases} \tilde{f}_k(s) & \text{if } s \in [0, s_k], \\ (p|_{V_{k+1}})^{-1}(f(s)) & \text{if } s \in [s_k, s_{k+1}]. \end{cases}$$
 Note that $(p|_{V_{k+1}}) \circ \tilde{f}_k(s_k) = f(s_k) = (p|_{V_{k+1}}) \circ (p|_{V_{k+1}})^{-1}(f(s_k))$, so $\tilde{f}_k(s_k) = (p|_{V_{k+1}})^{-1}(f(s_k))$, so by Gluing lemma, $\tilde{f}_{k+1}$ is continuous. Note $p \circ \tilde{f}_{k+1} = f$ on $[0, s_{k+1}]$ and $\tilde{f}_{k+1}(0) = e_0$.

Now for uniqueness. Suppose another path $f' : I \rightarrow E$ contains e_0 and $p \circ f' = p \circ f = f$. Let k be minimal such that $f', \tilde{f}$ disagree on $[s_k, s_{k+1}]$. If $k > 0$, then $f', \tilde{f}$ agree on s_k . If $k = 0$, $f', \tilde{f}$ agree at 0. Either way, $f', \tilde{f}$ agree on s_k .

Since $p \circ f'([s_k, s_{k+1}]) = f([s_k, s_{k+1}]) \subset U_{k+1}$, $f'([s_k, s_{k+1}]) \subset p^{-1}(U_{k+1})$. Since $[s_k, s_{k+1}]$ is connected and $f'(s_k) = \tilde{f}(s_k)$, $f'([s_k, s_{k+1}]) \subset V_{k+1}$. Then we have $(p|_{V_{k+1}})(f'(s)) = (p|_{V_{k+1}})(\tilde{f}(s)) \rightarrow f'(s) = \tilde{f}(s)$ on $[s_k, s_{k+1}]$ (since $p|_{V_{k+1}}$ is a homeomorphism).

Let $p : E \rightarrow B$ be a covering map and $F : I^2 \rightarrow B$ continuous. Let $b_0 = F(0, 0)$ and $e_0 \in E$ such that $p(e_0) = b_0$. Then F lifts uniquely to $\tilde{F} : I^2 \rightarrow E$ with $\tilde{F}(0, 0) = e_0$.

Let $\gamma_1 : I \rightarrow B$ be given by $\gamma_1(s) = F(s, 0)$, so γ_1 is continuous. Similarly, let $\gamma_2 : I \rightarrow B$ be given by $\gamma_2(t) = F(0, t)$, which is also continuous. γ_1, γ_2 lift uniquely to $\tilde{\gamma}_1, \tilde{\gamma}_2 : I \rightarrow E$.

Let $\mathcal{U}$ be a cover of B with equally covered by p sets U . Since F is continuous, $\{F^{-1}(U)\}$ is an open cover of I^2 . Since I^2 is compact metrizable, by Lebesgue number lemma, pick a subdivision $0 = s_0 < s_1 < s_2 < \dots < s_n = 1$ and $0 = t_0 < t_1 < t_2 < \dots < t_m = 1$ of I^2 with $R_{ij} = [s_i, s_{i+1}] \times [t_j, t_{j+1}] \subset F^{-1}(U_{ij})$

for some $U_{ij} \in \mathcal{U}$. Enumerate $\{R_{ij}\}$ with $1, 2, \dots, N = nm$ by the map $(i, j) \mapsto (i + 1) + nj$. For $k \in [N]$, let x_k denote the bottom left corner of R_k and C_k the union of the left and bottom edges of R_k . We define $\tilde{F}$ inductively as follows.

Claim: For $k \in \{0\} \cup [N]$, there exists $F_k : (I \times \{0\}) \cup (\{0\} \times I) \cup \bigcup_{j=1}^k R_j \rightarrow E$ such that $p \circ \tilde{F}_k = F$ on the domain of $\tilde{F}_k$ and $\tilde{F}_k(0, 0) = e_0$.

$k = 0$ is the union of the paths $\tilde{\gamma}_1, \tilde{\gamma}_2$. Suppose F_k is defined. Note $C_{k+1} \subset \text{dom } \tilde{F}_k$. Pick V_{k+1} the slice of $p^{-1}(U_{k+1})$ containing $\tilde{F}(x_{k+1})$. Since $p \circ \tilde{F}_k(C_{k+1}) = F(C_{k+1}) \subset U_{k+1}$, we have $\tilde{F}_k(C_{k+1}) \subset p^{-1}(U_{k+1})$. Since $\tilde{F}_k(C_{k+1})$ is connected, $\tilde{F}_k(C_{k+1}) \subset V_{k+1}$. Note over $R_{k+1} \cap \text{dom } \tilde{F}_k = C_{k+1}$, $(p|_{V_{k+1}})^{-1} \circ F$ and $\tilde{F}_k$ have the same image under p . Since both these values are in V_{k+1} and V_{k+1} is mapped homeomorphically to U_{k+1} , we have $(p|_{V_{k+1}})^{-1} \circ F = \tilde{F}_k$ over C_{k+1} . By Gluing lemma, we can extend $\tilde{F}_k$ to $\tilde{F}_{k+1}$ by defining $\tilde{F}_{k+1} = (p|_{V_{k+1}})^{-1} \circ F$ over R_{k+1} . This completes the induction.

Now for uniqueness. Suppose $F', \tilde{F}$ are distinct lifts of F . Let $k \in [N] \cup \{0\}$ be minimal such that $F', \tilde{F}$ disagree over $D_k = (I \times \{0\}) \cup (\{0\} \times I) \cup \bigcup_{j=1}^k R_j$. The lifts $\tilde{\gamma}_1, \tilde{\gamma}_2$ are unique, so $k \neq 0$. Then $F', \tilde{F}$ agree at x_k . Note $\tilde{F}(R_k) \subset V_k$. Since $F'(R_k)$ is connected, $F'(R_k) \subset V_k$ as well. Also over R_k , $p \circ \tilde{F} = F = p \circ F'$. Since V_k is mapped homeomorphically to U_k by p , we have $F' = \tilde{F}$ over V_k .

Let $\mathcal{P}$ denote the set of paths in E starting at e_0 and ending in the fiber $p^{-1}(b_0)$ up to path homotopy. Define $\varphi : \pi_1(B, b_0) \rightarrow \mathcal{P}$ by $\varphi(\gamma) = \tilde{\gamma}$. Define the product $*$ on $\mathcal{P}$ by $\tilde{\gamma}_1 * \tilde{\gamma}_2 = \varphi(\gamma_1 * \gamma_2)$.

By construction, $\varphi(\gamma_1) * \varphi(\gamma_2) = \varphi(\gamma_1 * \gamma_2)$. $\tilde{\gamma} * e_{e_0} = \varphi(\gamma * e_{b_0}) = \varphi(\gamma) = \tilde{\gamma}$ and similarly $e_{e_0} * \tilde{\gamma} = \tilde{\gamma}$. $(\varphi(\gamma_1) * \varphi(\gamma_2)) * \varphi(\gamma_3) = \varphi(\gamma_1 * \gamma_2) * \varphi(\gamma_3) = \varphi((\gamma_1 * \gamma_2) * \gamma_3) = \varphi(\gamma_1 * (\gamma_2 * \gamma_3)) = \varphi(\gamma_1) * \varphi(\gamma_2 * \gamma_3) = \varphi(\gamma_1) * (\varphi(\gamma_2) * \varphi(\gamma_3))$, so $*$ is associative over $\mathcal{P}$. $\varphi(\gamma) * \varphi(\bar{\gamma}) = \varphi(\bar{\gamma}) * \varphi(\gamma) = \varphi(e_{b_0}) = e_{e_0}$, so $\mathcal{P}$ has inverses. $\mathcal{P}$ is a group.

If $\varphi(\gamma) \simeq_p \varphi(\gamma')$, then $p \circ \varphi(\gamma) \simeq_p p \circ \varphi(\gamma') \rightarrow \gamma \simeq_p \gamma'$, so φ is injective.

φ is surjective by construction (take $p \circ \tilde{\gamma}$), so φ is an isomorphism of groups.

Brouwer fixed point. Suppose FTSOC $f(x) \neq x$ for all $x \in B^2$. Let $g : B^2 \rightarrow B^2 - 0$ be given by $g(x) = \frac{1}{2}(f(x) - x)$ (g is well-defined since B^2 is convex). Let $g' = g|_{S^1}$, so g' is null-homotopic. Let $j : S^1 \rightarrow B^2 - 0$ be inclusion and $j' : S^1 \rightarrow B^2 - 0$ be $j'(x) = -x$.

I claim g' is homotopic to j' . Let $H : S^1 \times I \rightarrow B^2 - 0$ be given by $H(x, t) = \frac{1}{2}((1-t)f(x) - (1+t)x)$. Note $H(x, 0) = \frac{1}{2}(f(x) - x) = g'(x)$ and $H(x, 1) = -x = j'(x)$. Suppose FTSOC for $t \in (0, 1)$ that $(1-t)f(x) = (1+t)x$, so $f(x) = \frac{1+t}{1-t}x$, so $\|f(x)\| = \frac{1+t}{1-t} > 1$, absurd. Further, since B^2 is convex, the image of H lies in B^2 , so H is well-defined and continuous, ie $g' \simeq j'$.

I claim $j' \simeq j$. Let $\gamma : I \rightarrow \mathbb{R}$ be the path $\gamma(s) = s$ for all $s \in I$. Let $\gamma' : I \rightarrow \mathbb{R}$ be the path $\gamma'(s) = s + 1/2$. γ, γ' are homotopic by the homotopy $G : I^2 \rightarrow \mathbb{R}$ by $G(s, t) = s + t/2$. The continuous function $p : \mathbb{R} \rightarrow S^1$ given by $p(x) = (\cos 2\pi x, \sin 2\pi x)$ induces a homotopy $p \circ G$ of $p \circ \gamma$ with $p \circ \gamma'$. Let $q : I^2 \rightarrow S^1 \times I$ be the quotient map given by $q = (p|I) \times \text{id}_I$. Note $p \circ G$ is constant on the fibers of q : $p(0 + t/2) = p(1 + t/2)$, so q induces the continuous function $J : S^1 \times I \rightarrow B^2 - 0$ such that $J \circ q = p \circ G$. That is, $J(p(s), t) = p(G(s, t)) = p(s + t/2)$. At $t = 0$, $J(p(s), 0) = p(s)$, so $J(x, 0) = x = j(x)$ for all $x \in S^1$ and at $t = 1$, $J(p(s), 1) = p(s + 1/2) = -p(s)$, $J(x, 1) = -x = j'(x)$. Hence, J is a homotopy of j with j' .

Since $g' \simeq j'$ and $j' \simeq j$, we have $g' \simeq j$, absurd since j is not nullhomotopic, but g' is.

Munkres proof: Suppose FTSOC $f(x) \neq x$ for all $x \in B^2$. Then $f(x) - x$ is a nonvanishing vector field on B^2 , so for some $x_0 \in S^1$ we have $f(x_0) - x_0 = ax_0$ for some $a > 0$. Then $f(x_0) = (1+a)x_0 \rightarrow \|f(x_0)\| = 1+a > 1$, contradiction.

Let A be a matrix with positive entries. Then A has a positive eigenvalue.

Let $B = S^2 \cap \mathbb{R}_{\geq 0}^3$. Let $r : \mathbb{R}_{\geq 0}^3 - 0 \rightarrow B$ be given by $r(x) = x/\|x\|$. Treat A as a linear map $\mathcal{A} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$. Since the entries of A are positive, we may restrict the domain of $\mathcal{A}$ to B and the range to $\mathbb{R}_{\geq 0}^3 - 0$. Call this map $\mathcal{A}'$. Let $f = r \circ \mathcal{A}' : B \rightarrow B$.

Notice that $B \cong B^2$. I claim in general if $X \xrightarrow{h} B^2$, then continuous $f : X \rightarrow X$ has a fixed point. Note that $h \circ f \circ h^{-1} : B^2 \rightarrow B^2$ is continuous, so by Brouwer fixed point, $h \circ f \circ h^{-1}(b) = b$ for some $b \in B^2$. Then $f(h^{-1}(b)) = h^{-1}(b)$, ie $h^{-1}(b)$ is a fixed point of f .

Hence, we have $r(\mathcal{A}'(v)) = v$, so $\mathcal{A}'(v)$ and v are proportional. Since $v \in B$, $\mathcal{A}'(v) = Av = \lambda v$ for some $\lambda > 0$.

2 Strong Deformation Retracts and Homotopy Type

Let $h : (X, x_0) \rightarrow (Y, y_0), k : (X, x_0) \rightarrow (Y, y_1)$ be homotopic by $H : X \times I \rightarrow Y$. Then there exists a path $\alpha : I \rightarrow Y$ such that $k_* = \hat{\alpha} \circ h_*$. In fact, $\alpha(s) = H(x_0, s)$.

Proof: Let $[\gamma] \in \pi_1(X, x_0)$. It suffices to show $k \circ \gamma \simeq_p \bar{\alpha} * (h \circ \gamma) * \alpha$. Indeed, let $H' : I \times I \rightarrow Y$ be given by

$$H'(s, t) = \begin{cases} H(x_0, 1 - (1 - t)3s) & \text{if } s \in [0, 1/3], \\ H(\gamma(3s - 1), t) & \text{if } s \in [1/3, 2/3], \\ H(x_0, t + (1 - t)(3s - 2)) & \text{if } s \in [2/3, 1]. \end{cases}$$

Now suppose $f : X \rightarrow X$ is homotopic to id_X . Let $x_0 \in X, x_1 = f(x_0)$. Let $\alpha : I \rightarrow X$ be the path induced by H , which starts at x_0 and ends at x_1 . Then $f_* = \hat{\alpha} \circ \text{id}_{X*} = \hat{\alpha}$.

Suppose X, Y have the same homotopy type. Let $f : (X, x_0) \rightarrow (Y, y_0)$ and $g : (Y, y_0) \rightarrow (X, x_1)$ have the nice properties. Suppose $f_*([\gamma]) = f_*([\gamma'])$. Then $g_* \circ f_*([\gamma]) = g_* \circ f_*([\gamma']) \rightarrow \hat{\alpha}([\gamma]) = \hat{\alpha}([\gamma']) \rightarrow [\gamma] = [\gamma']$, since $\hat{\alpha}$ is a bijection, so f_* is injective.

Since $g_* \circ f_* = \hat{\alpha}$ and $\hat{\alpha}$ is surjective, g_* is surjective. By a similar argument, g_* is injective, so g_* is an isomorphism. It follows $f_* = (g_*)^{-1} \circ \hat{\alpha}$ is an isomorphism, so $\pi_1(X, x_0) \cong \pi_1(Y, y_0)$.

Munkres proof. Start with the elementary. Let $\beta_0, \beta_1, \gamma_0, \gamma_1 : I \rightarrow I^2$ be given by $\beta_0(s) = (s, 0), \beta_1(s) = (s, 1), \gamma_0(t) = (0, t)$, and $\gamma_1(t) = (1, t)$. Since I^2 is convex, $\beta_0 * \gamma_1 \simeq_p \gamma_0 * \beta_1$ by a path homotopy G .

Let $F : I^2 \rightarrow X \times I$ be given by $F(s, t) = (\gamma(s), t)$, $c, f_0, f_1 : I \rightarrow X \times I$ be the paths $c(t) = (x_0, t), f_0(s) = (\gamma(s), 0), f_1(s) = (\gamma(s), 1)$. Then $F \circ G$ is a path homotopy between $f_0 * c \simeq_p c * f_1$.

Let $H : X \times I \rightarrow Y$ be the path homotopy as before. Note $\alpha = H \circ c, h \circ \gamma = H \circ f_0, k \circ \gamma = H \circ f_1$. Then $H \circ F \circ G$ is a path homotopy between $(h \circ \gamma) * \alpha \simeq_p \alpha * (k \circ \gamma)$.

3 The Fundamental Group of S^n

Let $U, V \subset X$ be open such that $U \cup V = X, U \cap V \neq \emptyset$ is path connected. Let $x_0 \in U \cap V$. Let $i : U \rightarrow X, j : V \rightarrow X$ be the inclusion maps. Then

$$\pi_1(X, x_0) = \langle i_*(\pi_1(U, x_0)), j_*(\pi_1(V, x_0)) \rangle.$$

Let $[\gamma] \in \pi_1(X, x_0)$. Note that $I = \gamma^{-1}(X) = \gamma^{-1}(U \cup V) = \gamma^{-1}(U) \cup \gamma^{-1}(V)$. Let $\mathcal{A}_U$ denote the connected components of $\gamma^{-1}(U)$ and define $\mathcal{A}_V$ similarly. Then $\mathcal{A} = \mathcal{A}_U \cup \mathcal{A}_V$ is an open covering of I .

By Lebesgue number lemma, pick $0 = r_0 < r_1 < r_2 < \dots < r_n = 1$ such that $[r_k, r_{k+1}] \subset A$ for some $A \in \mathcal{A}$ for $k = 0, 1, 2, \dots, n-1$. If for $k \in [n-1]$ we have $[r_{k-1}, r_k], [r_k, r_{k+1}]$ are both contained in elements of $\mathcal{A}_U$ or both in $\mathcal{A}_V$, replace $[r_{k-1}, r_k] \cup [r_k, r_{k+1}]$ with $[r_{k-1}, r_{k+1}]$. Repeating this process, we have $0 = s_0 < s_1 < s_2 < \dots < s_m = 1$ such that $\gamma([s_k, s_{k+1}])$ alternates between containment in U and V . Hence, $\gamma(s_1), \gamma(s_2), \dots, \gamma(s_{m-1}) \in U \cap V$. By assumption, $\gamma(s_0) = \gamma(s_m) = x_0 \in U \cap V$.

Since $U \cap V$ is path connected, for $k \in [m-1]$ let δ_k be a path from x_0 to $\gamma(s_k)$. Let δ_0, δ_m be the trivial path from x_0 to x_0 . For notation, let $I_k = [s_k, s_{k+1}]$ and $\gamma_k = \delta_k * (\gamma|_{I_k}) * \overline{\delta_{k+1}}$ for $k \in \{0\} \cup [m-1]$. Then

$$\begin{aligned} \gamma &= \delta_0 * (\gamma|_{I_0}) * (\gamma|_{I_1}) * \dots * (\gamma|_{I_{m-1}}) * \overline{\delta_m} \\ &= (\delta_0 * (\gamma|_{I_0}) * \overline{\delta_1}) * (\delta_1 * (\gamma|_{I_1}) * \overline{\delta_2}) * \dots * (\delta_{m-1} * (\gamma|_{I_m}) * \overline{\delta_m}) \\ &= \gamma_0 * \gamma_1 * \dots * \gamma_{m-1}. \end{aligned}$$

Since γ_k is either a loop in U or V through x_0 , the result follows (the other direction is obvious).

The map $S^n - p \rightarrow \mathbb{R}^n$ given by $(x_1, x_2, \dots, x_n, x_{n+1}) \mapsto \left(\frac{x_1}{1-x_{n+1}}, \frac{x_2}{1-x_{n+1}}, \dots, \frac{x_n}{1-x_{n+1}} \right)$ is well-defined continuous. Let $f(y_1, y_2, \dots, y_n) = y_1^2 + y_2^2 + \dots + y_n^2$. The reverse map $\mathbb{R}^n \rightarrow S^n$ given by $(y_1, y_2, \dots, y_n) \mapsto \left(\frac{2y_1}{f(y_i)+1}, \frac{2y_2}{f(y_i)+1}, \dots, \frac{2y_n}{f(y_i)+1}, \frac{f(y_i)-1}{f(y_i)+1} \right)$ is also well-defined continuous. It can be shown that these maps are inverses of each other. Hence, $S^n - p \cong \mathbb{R}^n$ is simply connected. The space $U \cap V = S^n - p - q$ is homeomorphic to $\mathbb{R}^n - 0$, which is path connected. Hence, $S^n = U \cup V$ is simply connected.